On the set function in $\mathbb{R}^n$

measure and integral

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In $\mathbb{R}^n$, let u be a set function on $\mathbb{R}^n$ which is a non-negative function on a subcollection of the collection of all subsets of $\mathbb{R}^n$, i.e. $u : D \to [0, +\infty]$, $D \subseteq \mathcal{P}(\mathbb{R}^n)$.

We will construct the smallest set of assumptions on u such that the following limit converges.

$$Du(x) = \lim_{B \searrow x} \frac{u(B)}{|B|}$$

where $B \searrow x$ denotes the construction of B into a singleton x, and $|B|$ denotes the Lebesgue measure of B.

Assumption 1 (Measurable of balls) $\overline{\mathcal{B}_r(x)} \in D$ *for all* $x \in \mathbb{R}^n$ *and* $r > 0$

Consider a simpler version where B is a closed ball of radius $r > 0$ centered at x, i.e. $B = \overline{\mathcal{B}_r(x)}$. Define lim sup, lim inf as follows

- $\overline{D}u(x) = \limsup_{B \searrow x} \frac{u(B)}{|B|} = \lim_{r \to 0^+} \left(\sup_{t \in (0,r)} \frac{u(B_t(x))}{|B_t(x)|} \right)$
- $\underline{D}u(x) = \liminf_{B \searrow x} \frac{u(B)}{|B|} = \lim_{r \to 0^+} \left(\inf_{t \in (0,r)} \frac{u(B_t(x))}{|B_t(x)|} \right)$

The condition for the existence of $Du(x)$ is when $\overline{D}u(x) = \underline{D}u(x)$. If $\overline{D}u(x) > \underline{D}u(x)$, let $\alpha, \beta \in \mathbb{Q}$ such that $\overline{D}u(x) > \alpha > \beta > \underline{D}u(x)$. Then $Du(x)$ exists for almost every $x \in \mathbb{R}^n$ is true if $A_{\alpha\beta} = \{x \in \mathbb{R}^n : \overline{D}u(x) > \alpha > \beta > \underline{D}u(x)\}$ has measure zero for all $\alpha, \beta \in \mathbb{Q}$

If $x \in A_{\alpha\beta}$, then given any $\delta > 0$, there exist balls $\overline{B}, \underline{B}$ centered at x with radius less than δ such that $\frac{u(\overline{B})}{|\overline{B}|} > \alpha > \beta > \frac{u(\underline{B})}{|\underline{B}|}$. That is, the collections $\{\overline{B}\}$ and $\{\underline{B}\}$ cover $A_{\alpha\beta}$ finely.

Lemma 1 (Vitali Covering Theorem for Lebesgue measure) *A collection $\mathcal{V}$ of closed balls is said to cover a set E finely if for every $x \in E$ and $\eta > 0$, there is a ball B containing x whose radius is less than η.*

Suppose E is covered finely by a collection $\mathcal{V}$ of balls such that $0 < \mu_e(E) < +\infty$. Then, given any $\epsilon > 0$, there is a countable collection M of disjoint balls such that

$$\mu\left(E \setminus \bigcup_{B_k \in M} B_k\right) = 0 \text{ and } \sum_{B_k \in M} \mu(B_k) < (1+\epsilon)\mu_e(E)$$

Claim 1 *Given any open set U contains a subset $E \subseteq A_{\alpha\beta}$, the $u(U) \geq \alpha|E|_e$*

Given any subset $E \subseteq A_{\alpha\beta}$, there exists an open set U containing E. Since U is open, for each $x \in E$, we can find a collection of balls $\{\overline{B}\}$ that finely covers E and is contained in U. By Vitali Covering Theorem for Lebesgue measure on E with that collection of balls, given any $\eta > 0$, there exists a finite collection of pairwise disjoint balls $\{B_i\}_{i=1}^N$ contained in U such that $\left|E \setminus \bigcup_{i=1}^N B_i\right|_e < \eta/\alpha$. [1]

Assumption 2 (Additivity) $A, B \subseteq \mathbb{R}^n$ *and* $A \cap B = \varnothing \implies u(A \cup B) = u(A) + u(B)$

Corollary 1 (Subset) $A \subseteq B \subseteq \mathbb{R}^n \implies u(A) \leq u(B)$

[1] α positive since it is strictly larger than a non-negative term $u(\underline{B})$

Then,

$$
\begin{aligned}
u(U) &\geq u\left(\bigcup_{i=1}^{N} B_i\right) && \text{(subset)}\\
&= \sum_{i=1}^{N} u(B_i) && \text{(additivity)}\\
&> \alpha \sum_{i=1}^{N} |B_i| \\
&= \alpha \left|\bigcup_{i=1}^{N} B_i\right| && \text{(additivity of Lebesgue measure)}\\
&= \alpha \left|E \cap \bigcup_{i=1}^{N} B_i\right|_e && \text{(subset of Lebesgue measure)}\\
&= \alpha \left(|E|_e - \left|E \setminus \bigcup_{i=1}^{N} B_i\right|_e\right) && \text{(Caratheodory condition)}\\
&= \alpha |E|_e - \eta
\end{aligned}
$$

As the inequality is true for all $\eta > 0$, $u(U) \geq \alpha |E|_e$

By Vitali Covering Theorem for Lebesgue measure on $A_{\alpha\beta}$ with $\{\underline{B}\}$, given any $\epsilon > 0$, there exists a finite collection of pairwise disjoint balls $\{B_i\}_{i=1}^{N}$ such that $\left|A_{\alpha\beta} \setminus \bigcup_{i=1}^{N} B_i\right| < \epsilon$ and $\sum_{i=1}^{N} |B_i| < |A_{\alpha\beta}|_e + \epsilon$

Then

$$
\begin{aligned}
|A_{\alpha\beta}| + \epsilon &> \sum_{i=1}^{N} |B_i| \\
&> \frac{1}{\beta} \sum_{i=1}^{N} \mu(B_i) \\
&\geq \frac{1}{\beta} \sum_{i=1}^{N} \mu(B_i^o) && \text{(subset}^2\text{)}\\
&\geq \frac{\alpha}{\beta} \sum_{i=1}^{N} |B_i^o \cap A_{\alpha\beta}|_e && \text{(by claim 1)}\\
&\geq \frac{\alpha}{\beta} \left|A_{\alpha\beta} \cap \bigcup_{i=1}^{N} B_i^o\right|_e && \text{(subadditivity of Lebesgue outer measure)}\\
&\geq \frac{\alpha}{\beta} \left|A_{\alpha\beta} \cap \bigcup_{i=1}^{N} B_i\right|_e && (|A \cap B|_e \leq |A \cap B^o|_e + |A \cap B^b|_e \ ^3)\\
&\geq \frac{\alpha}{\beta} \left(|A_{\alpha\beta}|_e - \left|A_{\alpha\beta} \setminus \bigcup_{i=1}^{N} B_i\right|_e\right) && \text{(Caratheodory condition)}\\
&> \frac{\alpha}{\beta} |A_{\alpha\beta}|_e - \frac{\alpha}{\beta}\epsilon
\end{aligned}
$$

Send ϵ to 0^+ implies $|A_{\alpha\beta}| \geq \frac{\alpha}{\beta}|A_{\alpha\beta}|$. Hence $|A_{\alpha\beta}| = 0$

Theorem 1 *Given a set function u on $\mathbb{R}^n$ that is (1) defined for all closed balls $\overline{\mathcal{B}_r(x)}, r > 0$ with (2) additivity, i.e. $A, B \subseteq \mathbb{R}^n$ and $A \cap B = \varnothing \implies u(A \cup B) = u(A) + u(B)$, Then*

$$Du(x) = \lim_{B \searrow x} \frac{u(B)}{|B|}$$

is defined for almost every $x \in \mathbb{R}^n$ and $B \searrow x$ denotes the contraction of closed balls $\overline{\mathcal{B}_r(x)}, r > 0$ into a singleton x

Remark 1 *Some remarks*

- $Du(x) = 1$ *everywhere for* u *being the Lebesgue measure.*
- $Du(x)$ *is defined everywhere for* u *being a measure on* $\mathbb{R}^n$.
- *(Lebesgue Differentiation Theorem) If* $f : \mathbb{R}^n \to [0, +\infty]$ *is Lebesgue measurable, let* $u(E) = \int_E f$.